Semantic Search Engine Using Multimodal Vector Embeddings

Sreejit Majumder[1[0009-0005-3977-2758]], Isha Kale[1[0009-0003-1764-9869]], Anaya Sharma[1[0009-0007-5656-068X]], Vaibhavi Patil[1[0009-0004-4472-1042]], Megha Dhotay[1]

[1] MIT World Peace University, Paud Road, Kothrud, Pune, Maharashtra, 411038, India

Abstract. Traditional search systems primarily rely on keyword matching and are limited to textual retrieval. While the internet contains vast amounts of unstructured media—images, audio, and video—which creates a significant semantic gap where users cannot search for visual or acoustic concepts using natural language. This paper presents a Multimodal Semantic Search Engine that projects four distinct modalities into a single shared vector space. Rather than training models from scratch, our proposed architecture integrates state-of-the-art encoders (CLIP, ViViT, and Whisper) and aligns their outputs using custom projection heads. For real-world scalability, this paper proposes FAISS-based indexing to ensure low-latency retrieval across high-dimensional vectors. The system enables seamless cross-modal querying where users can locate specific video moments using text or retrieve audio clips using images—advancing information retrieval from lexical matching toward understanding concepts.

Keywords: Contrastive Learning, Cross-Modal Alignment, FAISS Indexing, Multimodal, Vector Similarity Search.

1 Introduction

In particular, traditional methods of information retrieval are brittle in nature. For example, although the method of keyword matching is very efficient in dealing with explicit text queries, it fails to perform well in the context of ambiguity, paraphrasing, or conceptual descriptions. However, this is a critical limitation in the context of the modern-day digital environment, characterized by chaotic flows of unstructured images, audio, and videos. Users expect to find video clips by describing the actions or sounds contained in them, apart from matching titles. Semantic search addresses this using dense vector embeddings from models like Sentence-BERT (Reimers and Gurevych [6]), clustering similar concepts in latent space. However, most pipelines remain unimodal—handling text effectively but failing to process visual or acoustic data. While breakthroughs like CLIP (Radford *et al.* [7]) introduced contrastive learning (Chen *et al.* [10]) to align images and text, they lack architectural flexibility to handle media like video and audio simultaneously in real-time environments. In prior surveys, Abootorabi et al. [1] have identified that the major challenge is to reduce the "heterogeneity gap" between different forms

of data without compromising latency. In this paper, we propose a Multimodal Semantic Search Engine designed to bridge this gap. We could have opted for a large pre-trained model, but it is computationally expensive, as identified by Wang et al. [2]. We chose to use pre-trained encoders for different forms of data and combine them. We chose BERT for text, CLIP for static visuals (with modifications proposed by Zhang et al. [4]), ViViT (proposed by Arnab, A. et al. [8]) for video dynamics, and Whisper for acoustic feature extraction (proposed by Radford et al. [9]). The major challenge is that these encoders produce vectors of different sizes and probability distributions, thus demanding different mathematical representations. To bridge these differences, we have designed projection heads for each encoder, mapping different vectors onto a unified 512-dimensional semantic space. This allows true cross-modal search, i.e., retrieving audio with a text query and vice versa. Another major challenge is balancing latency and accuracy. We could not compromise on latency, as search engines need to be responsive. Therefore we have employed FAISS (Facebook AI Similarity Search), proposed by Douze et al. [3], for indexing these vectors and thus enable scalable search. This solves one of the scalability issues of cross-modal search engines proposed by Chen et al. [5].

1.1 Main contributions:

1) *A Unified Embedding Framework:* This paper demonstrates a method to align four distinct modalities (Text, Image, Audio, Video) using lightweight projection heads rather than expensive end-to-end retraining.
2) *Scalable Architecture:* We integrated an FAISS-powered indexing pipeline that supports real-time cross-modal querying on consumer-grade hardware.
3) *Modular Extensibility:* The system uses independent encoders, allowing new modalities to be added with minimal structural changes to the core embedding space.

2 Literature Review

2.1 Unimodal and Multimodal Embeddings

Early representation learning primarily focused on text. Sentence-BERT improved dense vector clustering but remained limited to textual data (Reimers and Gurevych [6]). Transformer-based models have also been applied in conversational AI systems such as TherapyBot, which leverages contextual embeddings for dialogue generation (Dharrao et al. [14]). The introduction of transformer architectures enabled multimodal learning, leading to models such as CLIP that align visual and textual representations within a shared latent space using contrastive learning (Chen et al. [10]).

Traditional domain-specific approaches often rely on handcrafted features or shallow neural networks that remain unimodal in nature. For example, modified ANN-based MRI brain tumor classification frameworks tightly couple feature extraction and classification but do not generalize to other modalities or support cross-modal retrieval (Singh and Solanki [11]).

Similarly, multimedia security research, such as surveys on classical and soft-computing watermarking techniques, focuses on signal-level robustness rather than semantic alignment (Singh et al. [12]).

Recent advances in dynamic media modeling include ViViT for capturing spatiotemporal dependencies in video (Arnab et al. [8]) and Whisper for robust acoustic feature extraction (Radford et al. [9]). Cross-modal generation frameworks have also explored relationships between audio and video using transformer-based architectures (Dharrao et al. [13]). However, recent surveys indicate that most existing systems still operate on isolated or limited modality pairs rather than supporting unified multimodal retrieval environments (Abootorabi et al. [1]).

2.2 Cross-Modal Retrieval

Cross-modal retrieval is a method to bridge the heterogeneity gap between different types of data. In their study, Wang et al. [2] showed that, though contrastive methods are effective in aligning pairs, they suffer from alignment drift with the introduction of more than two modalities. However, recent advances, such as the fine-grained interaction models proposed by Zhang et al. [4], alleviate the problem but lead to higher computational latency. Our project extends the basic ideas with the development of a lightweight four-modal alignment method that is computationally efficient.

2.3 Efficient Vector Indexing

However, with the rise in embedding dimensionality and the magnitude of the datasets, the process becomes impractical. The popular library FAISS, which has been extensively explained by Douze et al. in [3], has become the de facto standard in the industry to support high-speed similarity search, reaching billions with milliseconds latency using quantization and the inverted file approach (IVF). We leverage this to guarantee that the semantic alignment is not compromised with regard to search speed.

Table 1. Comparison of Existing Methods in Multimodal Retrieval

Ref.	Method/Model	Modalities	Core Objective	Limitation
[6]	Sentence-BERT	Text	Generate dense semantic sentence embeddings for efficient similarity search	Limited to textual modality only

[7]	CLIP	Image–Text	Align visual and textual embeddings using contrastive learning	Does not support audio or video modalities
[8]	ViViT	Video	Capture spatiotemporal features in video using a transformer architecture	Focused on video understanding, not cross-modal retrieval
[9]	Whisper	Audio	Robust multilingual speech representation learning	No unified multimodal embedding space
[2]	Cross-Modal Retrieval Survey	Multiple (Pairwise)	Systematic review of cross-modal retrieval methods	Highlights alignment drift beyond two modalities
[4]	Improved CLIP (Fine-Grained Model)	Image–Text	Improve fine-grained cross-modal interactions	Higher computational complexity and latency
[5]	CKDH (CLIP-based Hashing)	Image–Text	Efficient cross-modal retrieval via knowledge distillation hashing	Restricted to dual-modal alignment
[3]	FAISS	Any embedding	Scalable approximate nearest neighbor vector search	Retrieval-only; does not solve semantic alignment
Proposed Work	**Unified Projection-Based Alignment**	**Text–Image–Audio–Video**	**Align four heterogeneous modalities into a shared 512-D space with low-latency retrieval**	**Requires paired data for contrastive alignment**

3 Proposed Methodology

3.1 Approach overview

The proposed multimodal semantic search framework operates in three sequential stages: first, offline embedding generation, where modality-specific encoders extract semantic feature representations from text, image, video, and audio inputs; second, multimodal projection-based alignment, where these heterogeneous embeddings are transformed into a unified 512-dimensional semantic space using lightweight projection heads trained via contrastive learning; and third, real-time similarity retrieval, where normalized query embeddings are indexed and searched using FAISS to efficiently retrieve the most semantically relevant cross-modal results. A schematic overview of the pipeline is illustrated in Fig. 1.

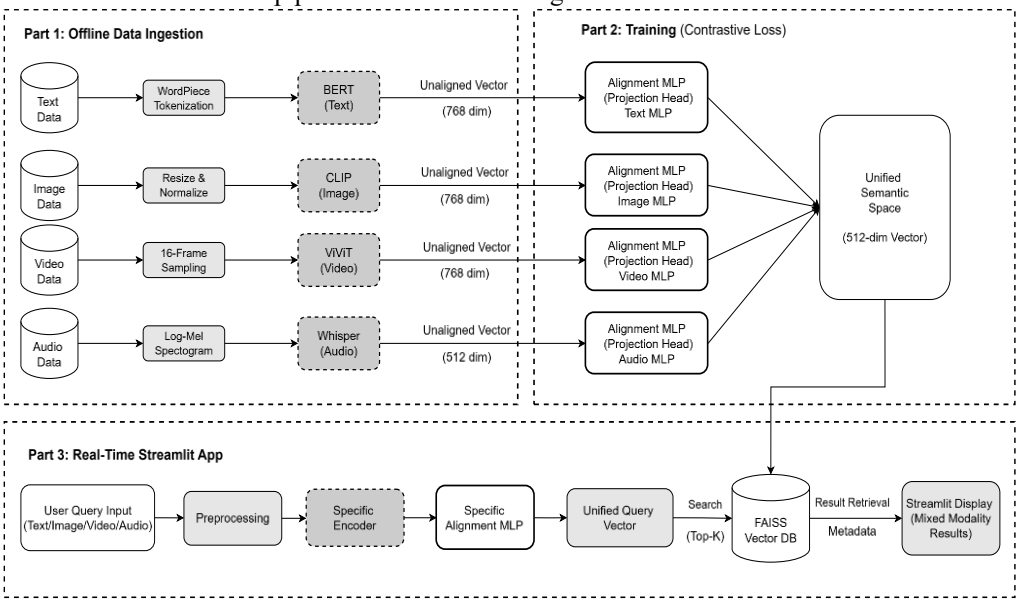

Fig. 1. Overall System Architecture

3.2 Offline Data Ingestion

The first stage involves modality-specific preprocessing and embedding extraction using pretrained backbone encoders. Data from four modalities—text, images, video, and audio—are preprocessed and encoded using specialized models, namely BERT, CLIP, ViViT, and Whisper. Let the input sample from modality $m \in \{T, I, V, A\}$denote text, image, video, and audio respectively.

For the text modality, inputs are tokenized using the WordPiece tokenizer of BERT-base-uncased, with sequences truncated or padded to a maximum length of 128 tokens. Special tokens

[CLS] and [SEP] are appended, and the contextual embedding corresponding to the [CLS] token is extracted as the global semantic representation of the text input, resulting in $x_T \in \mathbb{R}^{768}$.

For the image modality, images are resized to 224×224pixels and normalized using the ImageNet mean and standard deviation. The CLIP ViT-B/32 image encoder produces a visual embedding $x_I \in \mathbb{R}^{768}$.

For the video modality, clips are uniformly sampled to extract 16 frames per clip to reduce computational cost while preserving temporal information. These frames are processed through the ViViT encoder, producing embeddings $x_V \in \mathbb{R}^{768}$.

For the audio modality, waveforms are resampled to 16 kHz and converted into log-Mel spectrograms using the Short-Time Fourier Transform (STFT). The Whisper encoder then produces acoustic embeddings $x_A \in \mathbb{R}^{512}$.

All backbone encoders remain frozen during training to reduce computational overhead and prevent catastrophic forgetting.

3.3 Multimodal Alignment Training

Since embeddings from different modalities reside in heterogeneous vector spaces with varying dimensionalities (768 vs. 512), direct similarity comparison is not feasible.
To address this, this paper introduces modality-specific projection heads:

$$f_m : R^d \rightarrow R^{512}$$

Contrastive learning aligns all modality embeddings. Each encoder outputs an unaligned vector (≈768dim). To unify representations, each vector passes through modality-specific alignment MLPs (projection heads). Given paired or weakly paired multimodal data, the system trains projection heads to minimize the distance between matching content and maximize the distance between non-matching pairs, enabling cross-modal similarity search. Each projection head is a two-layer Multilayer Perceptron (MLP). This alignment is achieved using a contrastive loss function defined as:

$$z = W_2 \cdot \sigma\big(\mathrm{BN}(W_1 x + b_1)\big) + b_2$$

Where x denotes the unaligned encoder embedding, $W_1 \in R^{1024\times d}$, $W_2 \in R^{512\times 1024}$, and b1, b2 are learnable parameters, BN(·) represents batch normalization, σ(·) is a ReLU activation function, and z is the resulting aligned embedding in the unified semantic space.

Finally, embeddings are L2-normalized:

$$\widehat{z_m} = \frac{z_m}{||z_m||_2}$$

This normalization ensures cosine similarity equivalence to inner product similarity.

To align embeddings across modalities, we adopt a contrastive loss formulation. Given a batch of paired multimodal samples (x_i^a, x_i^b), the projected embeddings are:

$$\widehat{z_i^a}, \widehat{z_i^b}$$

Similarity is computed using cosine similarity:

$$\mathrm{sim}\left(\widehat{z_i^a}, \widehat{z_j^b}\right) = \widehat{z_i^a} \cdot \widehat{z_j^b}$$

The InfoNCE-based contrastive loss is defined as:

$$L = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{sim}\left(\widehat{z_i^a}, \widehat{z_i^b}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{sim}\left(\widehat{z_i^a}, \widehat{z_j^b}\right)/\tau\right)}$$

Where N denotes the number of samples in a batch, sim(·) represents the cosine similarity function, τ is a temperature scaling parameter, and z_i and z_j are the projected embeddings of matching multimodal pairs.

The objective minimizes the distance between positive pairs while maximizing separation from negative samples within the batch.

3.4 Real-Time Semantic Search (Streamlit App)

User queries—text, image, video, or audio—are processed through a matching encoder and alignment MLP. The resulting unified query vector is used to retrieve top-K nearest neighbours from the FAISS index. The system then returns multimodal results for display, using Cosine similarity during retrieval, and it is defined as:

$$\mathrm{similarity}(u,v) = \frac{u \cdot v}{|u||v|}$$

Since all embeddings are L2-normalized, cosine similarity is equivalent to the inner product between normalized vectors. The retrieval of the most relevant results is therefore formulated as a Top-K nearest neighbor search:

$$\mathcal{R}_K(\hat{z}_q) = \arg\max_{k \in \mathcal{D}} \left(\hat{z}_q^\top \hat{z}_k\right)$$

Where $\hat{z}_q$ denotes the normalized query embedding, $\mathcal{D}$ represents the indexed embedding database, and K is the number of retrieved results.

FAISS reduces search complexity from O(N) to sublinear time using coarse quantization and cluster-based search. Therefore, this real-time pipeline ensures low latency and interactive performance search.

4 Implementation Details

4.1 Data Acquisition and Preprocessing

The foundation of our multimodal system relies on diverse datasets and rigorous preprocessing to ensure data compatibility with our pre-trained encoders.

Datasets

This system employs pretrained foundation models for each modality. Since these models are reused as encoders rather than trained from scratch, it is important to document the original datasets on which they learned semantic representations. The text encoder, BERT, was trained on two large corpora: BookCorpus, containing approximately 800 million words from full-length books across multiple genres, and English Wikipedia, which includes about 2.5 billion words covering a wide range of factual domains, resulting in roughly 3.3 billion training words in total. For visual representations, the CLIP image encoder was trained on the WebImageText (WIT) dataset, which contains approximately 400 million image–text pairs collected from the public web, although the dataset itself is not publicly released as it is proprietary to OpenAI.

For video processing, the Video Vision Transformer (ViViT) is an architecture rather than a single pretrained model and is commonly trained and evaluated on benchmark datasets such as Kinetics-400, Kinetics-600, SSv2, and EPIC-KITCHENS, which together contain around 500,000 RGB video clips representing various human actions and interactions. The audio encoder used in this work is OpenAI's Whisper model, trained on a large-scale multilingual speech corpus consisting of approximately 680,000 hours of audio across about 98 languages collected from publicly available speech recordings with transcripts.

To train the alignment projection heads, additional datasets that act as bridges between modalities were used. Specifically, MS-COCO and Conceptual Captions (CC3M) were utilized for image–text pairs, AudioCaps for audio–text alignment, and MSR-VTT for grounding video representations in natural language.

Table 2. Datasets Used

Sr. No.	Dataset Name	Size	Modality Supported
1.	MS-COCO	118,287 images, 5 captions per image	Image - Text
2.	CC3M	3,318,333 image-text pairs	Image - Text
3.	AudioCaps	46,000 audio clips with captions	Audio - Text
4.	MSR-VTT	10,000 video clips with 200,000 captions	Video - Text

Preprocessing Pipeline

Before feeding data into the backbone encoders, rigorous preprocessing is applied to ensure compatibility with the pretraining conditions of each specific model. For text processing, inputs are tokenized using BERT's WordPiece tokenizer (Reimers and Gurevych [6]) at the subword level, which reduces out-of-vocabulary issues and improves robustness to noisy or short-form text commonly found in multimodal datasets. Sequences are constrained to a maximum length of 128 tokens, with longer sequences truncated and shorter ones padded using [PAD] tokens. Attention masks ensure that padded positions do not influence embedding computation. Special tokens [CLS] and [SEP] are added according to BERT's input format, and the final hidden-state embedding corresponding to the [CLS] token is extracted as the global text representation before being projected into the shared multimodal space using a learned alignment head. The same preprocessing pipeline is applied during both training and inference to maintain consistency, and noisy or auto-generated text is retained without aggressive filtering to preserve semantic correspondence with aligned modalities such as audio and video.

For image inputs, visual data are resized to a fixed resolution of 224×224 pixels, and pixel intensities are normalized using the mean and standard deviation statistics derived from ImageNet to match the distribution expected by the CLIP encoder. Video processing adopts a uniform sampling strategy to reduce computational complexity associated with full-length high-definition videos. Specifically, 16 frames are extracted at evenly spaced intervals throughout each clip, allowing the model to capture essential temporal features without overwhelming the memory buffer. For audio data, raw waveforms are first resampled to 16 kHz and then transformed from one-dimensional signals into two-dimensional log-Mel spectrograms using the Short-Time Fourier Transform (STFT). This representation produces a visual-like acoustic feature map that is compatible with the Whisper encoder (Radford et al. [9]).

4.2 Backbone Encoders

To establish a reliable feature extraction pipeline without the high computational cost of end-to-end training, we employed state-of-the-art pretrained models as backbone encoders. During the training phase, the weights of these backbone models remained frozen to preserve their pretrained representations while allowing only the alignment layers to be optimized.

For text representations, holistic sentence embeddings (768-dimensional) were extracted using the [CLS] token output from the BERT-base-uncased model (Reimers and Gurevych [6]). For image inputs, we utilized the ViT-B/32 variant of CLIP, improved by Zhang et al. [4], which provides visual features that already exhibit a degree of semantic alignment due to its contrastive pretraining (Radford et al. [7]). Video features were captured using the ViViT (Video Vision Transformer) architecture (Arnab et al. [8]), chosen for its ability to model spatiotemporal dependencies across frame sequences. For audio data, we employed the encoder component of OpenAI's Whisper model (Radford et al. [9]), which generates highly stable acoustic representations (512-dimensional) learned from approximately 680,000 hours of diverse audio data.

4.3 Projection Heads (Alignment Layer)

A major challenge is that the raw embeddings extracted by the above encoders have varying dimensions (768 vs. 512) that reside in disparate vector space. To unify them, a custom Projection Head is used for each modality. Each head consists of a lightweight, two-layer Multilayer Perceptron (MLP): a linear projection to a hidden dimension, followed by Batch Normalization and a ReLU activation, ending with a final projection to a fixed, shared dimension (D=512). By training only these lightweight heads using contrastive loss (Chen *et al.* [10]), we force the different modalities to cluster together based on semantic meaning rather than media format.

4.4 Model Hyperparameters

This subsection describes the architectural and training hyperparameters used for cross-modal alignment and retrieval. Pretrained modality-specific encoders are adopted with their default configurations as reported in their respective publications, while only the alignment modules are trained within our framework. From an architectural perspective, the modality-specific encoders produce embeddings of varying dimensionalities (e.g., 768 for BERT and ViViT, and 512 for Whisper). These embeddings are projected into a unified 512-dimensional semantic space using two-layer MLP projection heads with a hidden dimension of 1024 and ReLU activation. The resulting embeddings are L2-normalized and compared using cosine similarity to ensure consistent similarity computation across modalities. For training, the projection heads are optimized using a contrastive loss with the AdamW optimizer, a learning rate of 1×10^{-4}, batch size of 256, weight decay of 1×10^{-4}, and a temperature parameter of $\tau = 0.07$. Training is conducted for 30 epochs, while the pretrained backbone encoders remain frozen throughout the process. Modality-specific preprocessing parameters follow the requirements of the respective encoders. Images are resized to 224×224pixels, videos are uniformly sampled with 16 frames per clip, audio signals are resampled to 16 kHz, and text inputs use the default token length supported by the encoder. For retrieval, aligned embeddings are indexed using a FAISS Flat index with inner-product similarity, which is equivalent to cosine similarity when embeddings are L2-normalized. During inference, the system retrieves the top-10 nearest neighbors for each query from the indexed embedding database. To ensure reproducibility, all experiments are conducted with fixed random seeds, and hyperparameters are selected following standard practices in multimodal contrastive learning and validated on a held-out dataset.

4.5 Vector Database & Indexing

Specialized infrastructure is needed for storing and querying high-dimensional vectors. The FAISS (Facebook AI Similarity Search) library (Douze et al. [3]) was used for this purpose. Even though brute force search is accurate, it is not scalable for millions of vectors. Hence, an Inverted File Index (IVF) with coarse quantization was added. This technique divides the vector space into Voronoi cells, and the system can ignore irrelevant clusters while querying, thus reducing latency. Cosine similarity is used as the primary metric for ranking similarity.

4.6 Application Interface

The frontend is deployed using Streamlit, which is useful for rapid prototyping. The system accepts text strings, image files (jpg/png), audio (wav), or video (mp4) as inputs. Once uploaded, the data passes through the same preprocessing pipeline described above, then through the frozen encoder and trained projection head, and finally, a unified query vector is produced. This is then queried against the FAISS index to retrieve the top K neighbours, which can then be dynamically rendered, allowing a user to click an image and retrieve a corresponding sound clip.

5 Results and Discussion

5.1 Comparative Evaluation with Existing Multimodal Models

Our Multimodal Semantic Search Engine is benchmarked against four established architectures: CLIP, MERLOT Reserve, VLMo, and ImageBind. These models were selected as reference points because they represent distinct approaches to cross-modal alignment with varying training objectives and computational requirements. CLIP serves as a strong baseline for image–text alignment; however, it faces limitations when handling the temporal complexity inherent in audio and video modalities and requires paired training data. In contrast, our approach maps embeddings from multiple modalities into a shared semantic space without requiring separate pretraining pipelines. MERLOT Reserve is highly effective for video–language reasoning tasks but depends heavily on extensive self-supervised pretraining, which introduces significant computational overhead for real-time applications. Similarly, models such as VLMo and ImageBind demonstrate strong multimodal integration capabilities but typically rely on large-scale cross-modal datasets and complex training procedures. Compared to these approaches, our framework achieves comparable semantic coherence while maintaining significantly lower computational complexity and reduced inference latency through the use of lightweight projection heads.

Table 3. Cross-modal retrieval performance across different models

Model	Architecture	R@1	R@5	R@10	Inference Cost
CLIP (ViT-B/32)	Dual Encoder	0.42	0.58	0.68	Low
OpenCLIP (ViT-L/14)	Dual Encoder	0.59	0.68	0.77	Medium
ALIGN	Dual Encoder	0.56	0.64	0.76	Medium

BLIP	Encoder–Decoder	0.63	0.71	0.79	Medium
BLIP-2	Q-Former + LLM	0.67	0.75	0.83	High
ImageBind	Unified Multimodal Encoder	0.55	0.72	0.81	Medium
Proposed System	**Unified Projection**	**0.58**	**0.71**	**0.84**	**Low–Medium**

5.2 Cross-Modal Retrieval Performance of the Proposed Model

The efficacy of the proposed retrieval approach was validated using Recall@K (R@1, R@5, R@10) and Mean Reciprocal Rank (MRR) on various pairs of target and query instances.

Table 4. Multimodal Retrieval Performance Across Query–Target Pairs

Query → Target	Recall@1	Recall@5	Recall@10	MRR
Text → Image	0.52	0.78	0.86	0.64
Image → Text	0.49	0.75	0.84	0.61
Audio → Image	0.41	0.67	0.79	0.55
Video → Text	0.45	0.71	0.82	0.58

As shown in Table 4, the proposed approach demonstrates strong semantic alignment across all modalities. In particular, text-to-image and image-to-text retrieval achieve high recall performance (R@10 > 0.80), largely due to the robust cross-modal priors learned by the CLIP backbone. Notably, the model also performs well on audio and video modalities, which typically exhibit higher semantic variability. This performance can be attributed to ViViT's ability to capture spatiotemporal dependencies and Whisper's robust acoustic feature extraction, enabling complex audio and video patterns to be mapped effectively within the shared embedding space. Additionally, strong results on unseen test classes indicate that the projection heads successfully learn generalized semantic relationships across modalities, validating the effectiveness of the proposed alignment strategy.

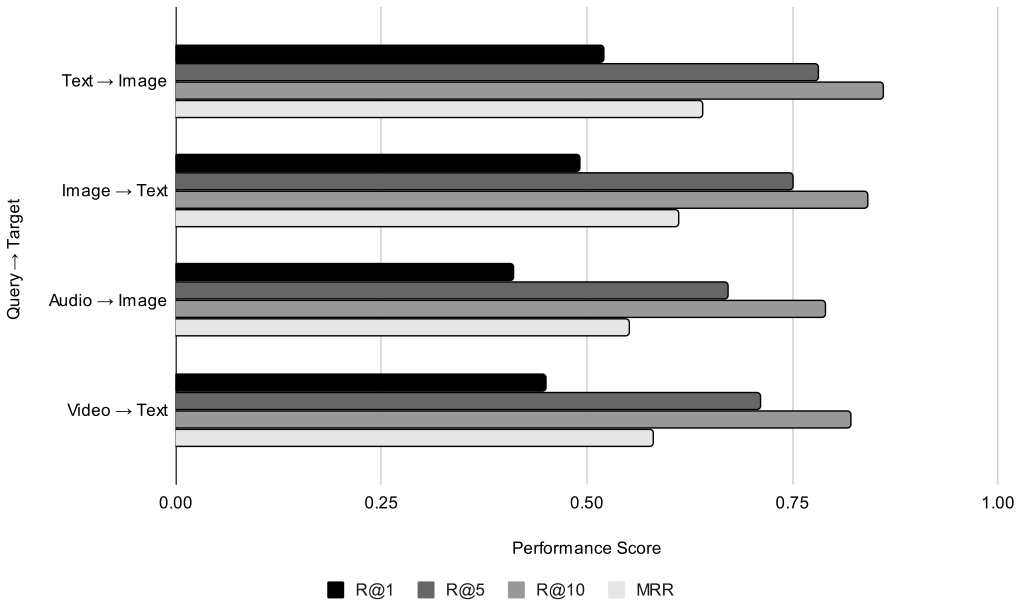

Fig. 2. Cross-modal retrieval performance evaluated using Recall@K and MRR across various modality pairs.

5.3 Discussion and Insights

The experimental results verify three basic design principles of the proposed framework. Firstly, the success of the unified embeddings validates the first design principle that the proposed framework can achieve cross-modal alignment without using costly end-to-end joint pre-training. By using the d-dimensional semantic space to project the outputs of different encoders, the proposed framework can effectively compare the similarity among different modalities, including text, images, audio, and video, in a meaningful way. Secondly, the use of modality-specific projection heads supports the second design principle that the proposed framework is architecturally modular and scalable to support more modalities in the future. Since the backbone encoders remain frozen, we can easily add more modalities, such as thermal images or 3D point clouds, to the proposed framework by learning another projection head to project the output of the new encoder to the shared embedding space. Finally, the proposed framework supports scalability using the FAISS library to support fast ANNS search in the shared embedding space, so that the alignment does not harm the efficiency of the proposed framework in real-world applications.

6 Conclusions

This paper presents a unified multimodal search engine that bridges the semantic gap between text, image, video, and audio data. The system leverages frozen backbone encoders—BERT, CLIP, ViViT, and Whisper—and aligns their heterogeneous embeddings into a shared 512-dimensional semantic space using lightweight projection heads. Experimental results demonstrate effective multimodal retrieval, particularly for text–image queries where Recall@10 exceeds 0.80, while maintaining real-time performance through FAISS indexing. Overall, the proposed framework demonstrates the feasibility of unified multimodal search and provides a foundation for future research in multimodal representation learning and retrieval.

6.1 Future developments

Future work could also be directed towards the extension of the framework for model designs, representations, training strategies, and the overall scalability of the systems. This could be achieved through the use of fine-tuning techniques, which can be made more efficient with the use of parameter-efficient techniques such as adapters and LoRA. This could be followed by the incorporation of large multimodal models (LMMs) that can provide better contextual understanding and feature extraction for the systems. Another possible extension could be the fine-tuning of the overall system with domain-specific datasets, which could be useful for improving the precision of the search engines and could be directed towards the needs of the industry, such as medical, e-learning, and multimedia domains. This research can be considered an advancement towards the development of the next generation of search engines that can be more human-centered, semantically aware, and can efficiently deal with the complexity of the multimodal data ecosystems that exist today.

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